

SEA SPIDERS

These fragile-looking marine animals live among seaweeds in shallow seas and coral reefs in tropical waters. The largest species live in the deep ocean.

Sea spiders of the class Pycnogonida are not true spiders, and are so remarkably different from other arthropods that some scientists think they belong to an ancient lineage that is not closely related to any groups alive today. Others think they are

distant cousins of the arachnids. Most sea spiders are small – less than a centimetre ($\frac{3}{8}$ in) in body length. They have three or four pairs of legs and the head and thorax are fused together. Instead of claw-like mouthparts they have a stabbing proboscis, which they use like a hypodermic needle to suck fluids from their invertebrate prey. Their spindly body shape means they do not need gills; they rely instead on oxygen seeping directly into all their cells through their body surface.

PHYLUM	ARTHROPODA
CLASS	PYCNOGONIDA
ORDER	1
FAMILIES	13
SPECIES	1,348



GIANT SEA SPIDER
Colossendeis megalonyx
F: Colossendeidae

One of the largest of all sea spiders, this is a deep-sea animal of sub-Antarctic waters, with a leg span of 70 cm (28 in).



SPINY SEA SPIDER
Endeis spinosa
F: Endeidae

This sea spider – found around European coasts but perhaps ranging elsewhere – has an especially spindly body, with a long cylindrical proboscis.



FAT SEA SPIDER
Pycnogonum littorale
F: Pycnogonidae

Unlike most other sea spiders, this European species has a thick body and comparatively short, curved, clawed legs. It feeds on anemones.

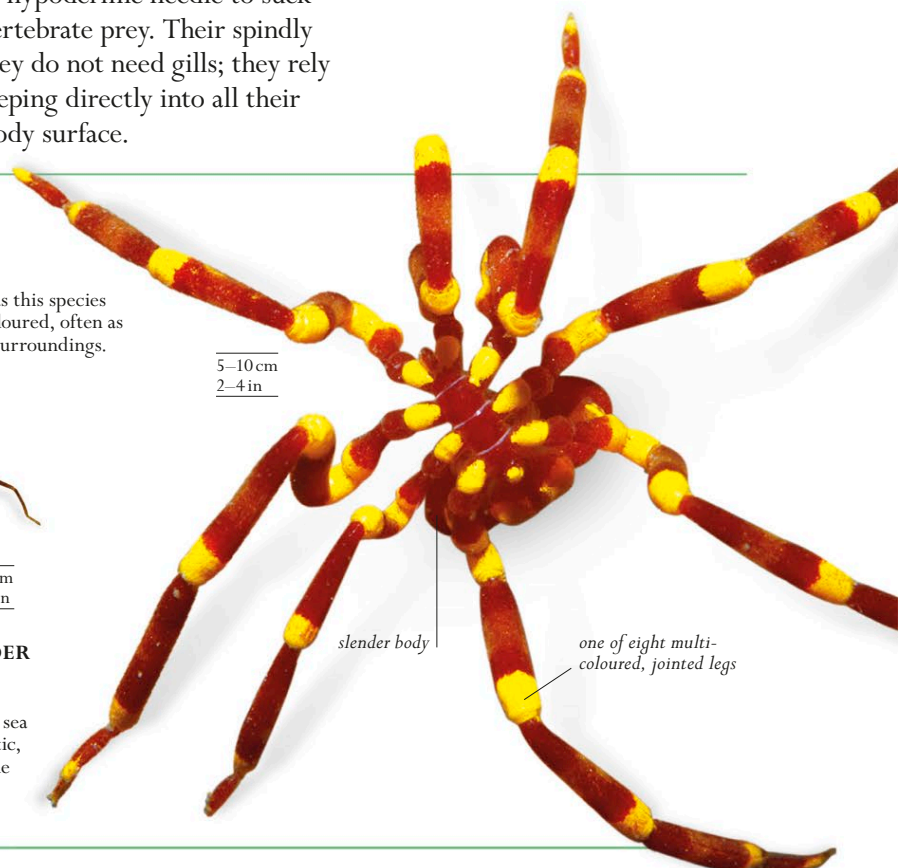
YELLOW-KNEED SEA SPIDER
Unknown sp.
F: Callipallenidae

Some species of sea spiders – such as this species of Australian reefs – are strikingly coloured, often as camouflage against their colourful surroundings.



GRACEFUL SEA SPIDER
Nymphon brevirostre
F: Nymphonidae

One of the most common sea spiders of the N.E. Atlantic, this species occurs in the intertidal zone and in offshore shallows.



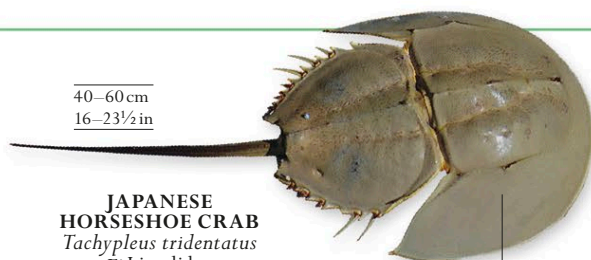
HORSESHOE CRABS

The Merostomata are a small class of sea-dwelling relatives of spiders and scorpions. Of prehistoric origin, they are widely regarded as “living fossils”.

Horseshoe crabs were far more diverse in prehistory, when animals like them may well have been the first chelicerates on Earth. Their claw-like mouthparts and lack of antennae indicate that, despite their hard carapace,

horseshoe crabs are most closely related to arachnids. The leaf-like gills on the underside of their abdomens are the forerunners of similar internal structures – the so-called book lungs – that arachnids use to breathe on land. The pedipalps are used as a fifth pair of legs, one pair more than in spiders. Horseshoe crabs grub around in muddy ocean waters to catch prey. They come ashore in large numbers to breed, laying their spawn in sand.

PHYLUM	ARTHROPODA
CLASS	MEROSTOMATA
ORDER	1
FAMILY	1
SPECIES	4



JAPANESE HORSESHOE CRAB
Tachypleus tridentatus
F: Limulidae

This horseshoe crab spawns in sandy shores of E. Asia. Habitat destruction and pollution have depleted its numbers in some parts of its range.

fused head and thorax covered by carapace

long spine-like tail

MANGROVE HORSESHOE CRAB
Carcinoscorpius rotundicauda
F: Limulidae

Associated with muddy-sand habitats in S.E. Asia, this species feeds on insect larvae and other invertebrates, as well as small fish.

